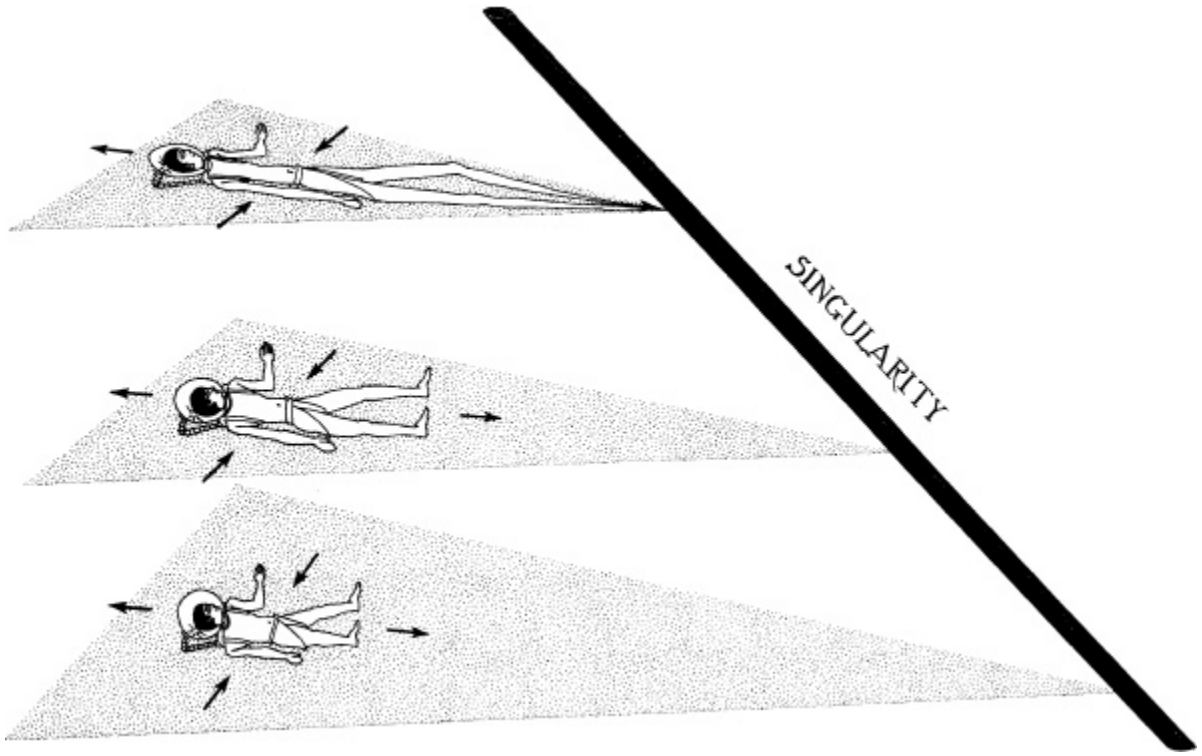


13.1 Spacetime diagram depicting the feet-first fall of an astronaut into the singularity at a black hole's center, according to the Oppenheimer–Snyder calculations. As in all previous spacetime diagrams (for example, Figure 6.7), one spatial dimension is missing; that is why the astronaut looks two-dimensional rather than three-dimensional. The singularity is tilted in this diagram, in contrast to its vertical position in Figure 6.7 and Box 12.1, because the time plotted upward and the space plotted horizontally here are different from there. Here they are the astronaut's own time and space; there they were Finkelstein's.

The mechanism of destruction is not fully clear in Figure 13.1, because the figure ignores the curvature of space. In fact, as the astronaut's body reaches the singularity, it gets stretched out to truly infinite length and squashed transversely to truly zero size. The extreme curvature of space near the singularity permits him to become infinitely long without shoving his head out through the hole's horizon. His head and feet are both pulled into the singularity, but they are pulled in infinitely far apart.

Not only is an astronaut stretched and squeezed infinitely at the singularity, according to the Oppenheimer–Snyder equations; all forms of matter are infinitely stretched and squeezed—even an individual atom; even the electrons, protons, and neutrons that make up atoms; even the quarks that make up protons and neutrons.

Is there any way for the astronaut to escape this infinite stretch and squeeze? No, not after he has crossed the horizon. Everywhere inside the horizon, according to the Oppenheimer–Snyder equations, gravity is so